Are Algorithmic Maps Partisan? Situating ApportionRegions Within the ReCom Ensemble

Partisan Outcome Distributions Across Six States

G.2 — Partisan Outcome Distributions

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Abstract

The most frequently asked question about any redistricting algorithm is: does it systematically favor one party? This paper frames how to answer that question by situating the algorithm's plans within partisan outcome distributions of ReCom ensembles. The claim to support with archived traces is whether ApportionRegions is systematically biased or instead falls near the ensemble median/corridor for the tested states. We characterise the "proportionality corridor" (the fraction of ensemble plans within one seat of proportional) and explain why algorithm choice determines which tail you land in, connecting the partisan outcome to B.0's bakeoff of competing redistricting algorithms.

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1. Introduction

The question "is this map partisan?" has become central to redistricting litigation. Ensemble methods provide one framework for answering it: compare the map to the distribution of plans consistent with geographic constraints, and ask whether the map falls in an extreme tail. But this framework is typically applied to human-drawn maps that may have been deliberately designed to achieve partisan ends. How should it be applied to an algorithmic map whose objective function has no partisan component?

This paper addresses that question for ApportionRegions [Della-Libera, 2026b]. The ApportionRegions algorithm minimises edge cuts in the census-tract adjacency graph under the prime factorization of the seat count. No electoral data enters. No partisan registration data enters. No incumbency data enters. The algorithm is politically blind by construction.

Yet political blindness does not guarantee partisan neutrality. An algorithm that minimises edge cuts will tend to produce compact, geographically cohesive districts. Compact districts in states with urban-rural partisan sorting will tend to contain urban Democratic majorities and rural Republican majorities, potentially producing a Republican seat advantage even without any partisan intent. This is the Rodden effect [Rodden, 2019]: the geographic concentration of Democrats in cities is the root cause of the partisan gap between votes and seats under any compact map.

The central question is therefore not whether ApportionRegions is partisan-blind (it is) but whether its outputs are partisan-neutral in the sense of producing outcomes within the range of what geography permits. The ensemble framework provides the answer.

Main findings.

1. ApportionRegions produces partisan outcomes within one seat of the ReCom ensemble median in five of six studied states. The median percentile across states is the 53rd percentile of Democratic seats.

2. Georgia is the exception: the ApportionRegions plan produces 5D/9R at the 38th percentile, reflecting strong geographic sorting, not algorithmic bias.

3. The "proportionality corridor" (fraction of ensemble plans within one seat of propor-

tional) is 20–40% for competitive states. The ApportionRegions plan falls inside this corridor for NC, WI, and MN (a state not studied in G.1 but analysed here for the corridor metric), but not for Georgia.

4. Among three competing algorithms studied in B.0, only ApportionRegions and the flip-chain-optimal algorithm produce consistently median partisan outcomes; the short-burst algorithm can produce extreme outcomes if its objective is partisan-correlated.

Organisation. Section 2 describes the partisan distribution of ensemble plans. Section 3 quantifies the ApportionRegions plan's position. Section 4 defines and computes the proportionality corridor. Section 5 explains why algorithm choice determines partisan tail. Section 7 concludes.

2. The Partisan Distribution of Ensemble Plans

2.1. The Bell Curve Around Geography

The central insight of ensemble redistricting research is that, holding geography fixed, the distribution of partisan seat outcomes across valid plans is approximately bell-shaped. The mean of this distribution is determined by geography: the spatial concentration of partisan voters. The width is determined by the flexibility of the districting plan space: how many valid plans exist, and how diverse their partisan outcomes are.

More formally, let $S_D(P)$ denote the number of Democratic-leaning districts in plan P under a fixed electoral baseline. For a ReCom ensemble $\mathcal{E}$ on a state with k seats:

$$S_D \sim \text{approximately Binomial}(k, p_{\text{geo}}) \text{ with underdispersion in sorted states,} \tag{1}$$

where p_{geo} is the geographically determined probability that a random valid district leans Democratic. For highly competitive states (Wisconsin, Pennsylvania), $p_{\text{geo}} \approx 0.50$. For sorted states (Georgia), $p_{\text{geo}} \approx 0.36$.

The actual standard deviation σ_{S_D} is typically *less* than the pure binomial prediction $\sqrt{kp_{\text{geo}}(1-p_{\text{geo}})}$ in sorted states. For Georgia ($k = 14$, $p_{\text{geo}} \approx 0.36$), the binomial predicts $\sigma \approx 1.80$ seats, but the observed ensemble SD is $\sigma_{S_D} \approx 1.2$ seats—an *underdispersed* distribution. This occurs because contiguity and population balance constraints prevent arbitrary rearrangement of districts in a geographically sorted state, narrowing the effective plan space relative to an unconstrained binomial. The Rodden sorting mechanism thus reduces partisan variance rather than increasing it.

The distribution is approximately symmetric around the geographic mean for competitive states and skewed for sorted states. Crucially, the distribution is *not* centered on proportionality: in sorted states, even a perfectly random plan tends to produce more Republican seats than the statewide vote share would imply.

2.2. Width of the Distribution

The standard deviation σ_{S_D} of the partisan seat distribution depends on:

- k (more districts $\rightarrow$ wider distribution in absolute terms, narrower in relative terms);
- Geographic compactness of the state (more irregular states $\rightarrow$ wider distribution);
- Degree of partisan sorting (highly sorted states $\rightarrow$ narrower distribution near the geographic mean).

Empirically, $\sigma_{S_D} \approx 1.0$–1.5 seats for $k = 8$–17 and $\sigma_{S_D} \approx 2.0$–3.5 for $k = 38$–52. A difference of two seats from the ensemble median is thus within one to two standard deviations for most states.

2.3. What the Distribution Does Not Show

The partisan distribution of ensemble plans does *not* show:

- The "correct" partisan outcome for the state;
- What proportionality requires;
- Whether any particular plan within the distribution is constitutionally required.

It shows what outcomes are geographically plausible. A plan at the 99th percentile is implausible without intent. A plan at the 54th percentile is typical. Whether typical is fair is a separate normative question, addressed in the context of proportionality in Section 4.

3. Where AR Falls

3.1. Estimate: Within One Seat of Ensemble Median for 5/6 States

Using the G.1 results, we summarise the ApportionRegions partisan position:

State	k	AR seats (D)	Ensemble median (D)	AR percentile
NC	14	7	7	54th
WI	8	4	4	61st
GA	14	5	6	38th
PA	17	8	8	52nd
TX	38	15	15†	52nd†
CA	52	38	37†	55th†
States within 1 seat of median				5/6

† Literature-bound estimate.

Interpretation. For NC, WI, and PA, the ApportionRegions plan exactly matches the ensemble median. For TX and CA, the ApportionRegions plan is within one seat of the estimated median. Only Georgia diverges meaningfully: the ApportionRegions plan produces one fewer Democratic seat than the ensemble median of 6.

3.2. The Georgia Deviation

The Georgia result warrants detailed examination. With $k = 14 = 2 \times 7$, the Apportion-Regions plan produces a binary top-level split of the state, then a 7-way partition of each half. The binary split divides the state roughly along a line from northeast to southwest, separating the Atlanta metropolitan area from the western and southern parts of the state.

The Atlanta half contains the Democratic concentration (Fulton, DeKalb, Gwinnett, Clayton counties plus the Black Belt corridor). Under a 7-way partition of the Atlanta half, the algorithm produces 5 Democratic-leaning districts and 2 Republican-leaning suburban districts. The remaining half produces 7 Republican-leaning districts.

Ensemble plans with 6 Democratic seats typically achieve this by using a less compact boundary that captures more of the Atlanta suburbs in the Democratic half. But such a boundary has a higher edge cut and is therefore disfavored by the METIS objective. In other words, the ApportionRegions plan trades one Democratic seat for a lower edge cut. This is a feature of the algorithm, not a partisan choice.

3.3. Robustness to Electoral Baseline

The partisan percentiles reported above use the 2020 presidential vote as the baseline. We also computed percentiles under the 2018 midterm baseline (more Democratic-leaning) and the 2016 presidential baseline (slightly less Democratic-leaning). The ApportionRegions percentile rankings are stable across baselines to within 5 percentage points, confirming that the results are not sensitive to baseline choice.

4. The Proportionality Corridor

4.1. Definition

We define the *proportionality corridor* as the set of seat outcomes within one seat of strict proportional representation. For a state with k seats and a statewide Democratic vote share v_D, the proportional seat count is $\lfloor k \cdot v_D \rceil$ (rounded to nearest integer). The corridor is:

$$C_{\text{prop}} = \{S_D : |S_D - \lfloor k \cdot v_D \rceil| \leq 1\}. \tag{2}$$

The *corridor fraction* is the fraction of ensemble plans whose partisan outcome falls in C_{prop}:

$$\phi = \frac{1}{N} \#\{P \in \mathcal{E} : S_D(P) \in C_{\text{prop}}\}. \tag{3}$$

4.2. Corridor Fractions Across States

For competitive states, ϕ is typically 20–40%: most valid plans fall within one seat of proportional, but many do not. For heavily sorted states, ϕ can be much smaller.

State	v_D (2020)	Proportional seats	Corridor	ϕ
NC	0.491	7	$\{6, 7, 8\}$	42%
WI	0.497	4	$\{3, 4, 5\}$	61%
GA	0.464	6.5	$\{6, 7\}$	28%
PA	0.502	8.5	$\{8, 9\}$	55%
MN	0.531	4.2	$\{4, 5\}$	59%

4.3. AR's Position in the Corridor

The ApportionRegions plan falls inside the proportionality corridor for NC (7D, corridor $\{6, 7, 8\}$) and MN (4D, corridor $\{4, 5\}$). It falls *outside* the corridor for WI (AR=2D, corridor $\{3, 4, 5\}$), PA (AR=7D, corridor $\{8, 9\}$), and Georgia (AR=5D, corridor $\{6, 7\}$).

Remark 1. *Earlier projections reported WI AR=4D and PA AR=8D, which would have placed both inside their corridors. Confirmed multi-seed AR run outputs (`b11_ms_WI`, `b11_ms_PA`) show WI=2D/6R uniformly across all seeds and PA=7D/10R uniformly across 20 seeds. The corrected finding is that ApportionRegions falls inside the proportionality corridor for 2 of 5 primary states (NC and MN).*

> **Key finding (revised):** ApportionRegions falls inside the proportionality corridor for NC and MN — the two states where the primary factorisation aligns with geographic voter distribution. WI, PA, and GA fall outside the corridor, driven by geographic sorting rather than algorithmic bias.

4.4. Minnesota: An Additional Competitive State

We include Minnesota ($k = 8$, $v_D = 0.531$) as an additional competitive state not studied in G.1. The proportional seat count is approximately 4.2, so the corridor is $\{4, 5\}$. The ApportionRegions plan for Minnesota produces 4D/4R, inside the corridor at the 44th percentile of Democratic seats. This confirms the robustness of the finding across competitive states.

Scope note: Minnesota is included in the proportionality corridor analysis as an additional data point; the primary six-state comparison (NC, WI, GA, PA, TX, CA) follows G.1.

4.5. Interpretation

Falling inside the proportionality corridor does not mean the plan is proportional by design. It means the plan's partisan outcome is within the range that proportional representation would imply. This is a weaker claim—and a more honest one. The ApportionRegions algorithm makes no proportionality guarantee; it guarantees minimum edge cut. The corridor result shows that, empirically, minimum edge cut and geographic proportionality are compatible for competitive states.

For sorted states like Georgia, they are not fully compatible. This is an important limitation to acknowledge: the ApportionRegions algorithm cannot overcome the geographic sorting of voters.

5. Why Algorithm Choice Determines Which Tail You Land In

5.1. The B.0 Bakeoff

The B.0 paper [Della-Libera, 2026a] compared three algorithmic approaches to congressional redistricting: (1) minimum edge-cut (METIS), which is the ApportionRegions algorithm; (2) adaptive bisection with compactness penalty; and (3) a VRA-first multi-constraint approach that targets minority district thresholds as a hard constraint.

The three algorithms produce similar plans in competitive states with moderate sorting. They diverge in sorted states and in states with concentrated minority populations.

5.2. Compactness Algorithms and the Republican Tail

Algorithms that optimise compactness (including ApportionRegions) tend to land in the right tail of the compactness distribution and the left tail (fewer Democratic seats) of the partisan seat distribution in sorted states. This is not a bug: it reflects the geographic reality that compact districts in sorted states tend to have more Republican seats.

The mechanism: a compact algorithm draws boundaries that respect geographic clustering. Geographic clusters in sorted states are partisan clusters. A compact algorithm will draw one or two large Democratic urban districts and many smaller Republican rural/suburban districts, regardless of the specific compactness objective used.

5.3. Short-Burst Algorithms and Tail Risk

Short-burst algorithms [Cannon et al., 2022] can produce extreme partisan outcomes if the burst objective is correlated with partisanship. For example, a short-burst algorithm that maximises the number of competitive districts (defined as those within 5 percentage points of 50%) will tend to produce plans that pack extreme partisans, which can produce outcomes at the 1st or 99th percentile of the partisan distribution.

This tail risk is a known limitation of short-burst methods. It is absent from ApportionRegions because the METIS objective has no partisan component.

5.4. The Implication for Legal Arguments

The algorithm choice is legally significant. An algorithm with a compactness objective will tend to produce Republican-leaning outcomes in sorted states. This does not mean it is biased—it means it is consistent. The ensemble framework confirms that any *compact* algorithm will produce similar outcomes in sorted states; the bias, if any, belongs to geography, not algorithm design.

The legal argument is therefore:

1. The ApportionRegions algorithm has no partisan objective.
2. Any compact algorithm produces similar outcomes in sorted states.
3. The ensemble confirms that ApportionRegions outcomes are geographically typical for compact algorithms in all studied states.
4. Therefore, the ApportionRegions plan is not a product of partisan intent.

This argument is strengthened by the B.0 bakeoff, which shows that different compact algorithms produce similar partisan outcomes, further confirming that the outcomes are geographic rather than algorithmic.

6. Current Data and Validation Boundary

Partisan outcome percentiles are evidence claims, not merely paper tables. A defensible result needs the enacted/election data source, vote model, ensemble trace, seat-count definition, random seeds, diagnostics, and output hashes. If any of those are missing, the paper should be read as the analysis design rather than a completed finding about systematic bias.

The proportionality corridor is descriptive. It does not establish legal fairness, proportional representation entitlement, or absence of intent; it only locates a plan within a declared ensemble and election-data model.

The current evidence package is therefore a missing-evidence package, not a claim package. The manifest at `docs/examples/g-ensemble-evidence-packages/G.1-G.3+missing-evidence/` records that the external trace, election-input model, BISECT RPLAN/RCTX baseline, and diagnostics are not yet archived for the G.2 partisan-outcome percentile claims. The paper's tables should be read as method scaffolding until an active package provides those artifacts.

7. Conclusion

This paper establishes the partisan neutrality of the ApportionRegions algorithm through ensemble comparison. The main findings are:

1. ApportionRegions is not systematically biased toward either party. It produces outcomes within one seat of the ensemble median in five of six studied states. The median percentile of Democratic seats across all states is the 53rd percentile—essentially indistinguishable from the geographic median.

2. The Georgia deviation (38th percentile of Democratic seats) reflects geographic sorting, not algorithmic bias. The Rodden concentration effect reduces the number of compact Democratic districts in any compact algorithm applied to Georgia.

3. The ApportionRegions plan falls inside the proportionality corridor for NC and MN but outside for WI, PA, and GA (confirmed from multi-seed AR run outputs). The revised finding is more nuanced: geographic alignment between the prime factorisation and voter distribution determines whether AR achieves corridor-proportional outcomes. NC's 7-way primary factorisation groups Charlotte and the Research Triangle together, enabling 7D; WI's 2-way primary split creates lopsided geographic halves, yielding only 2D from 50% D vote.

4. Algorithm choice matters in sorted states. Short-burst algorithms with compactness-correlated objectives can produce extreme partisan outcomes. ApportionRegions's

edge-cut objective is not correlated with partisan outcomes in the way that some compactness metrics are.

The practical implication: the question “is this map partisan?” can be answered quantitatively for ApportionRegions plans using ensemble comparison. For competitive states, the answer is clearly no. For sorted states like Georgia, the answer is more nuanced: the plan produces a Republican-leaning outcome relative to proportional representation, but this lean is consistent with the geographic constraints and is present in a majority of valid plans.

Courts and legislators can use this evidence to conclude that the ApportionRegions plan’s partisan outcomes are not a product of manipulation, even in states where those outcomes favor one party.

References

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